CSCI218: Foundations of Artificial Intelligence

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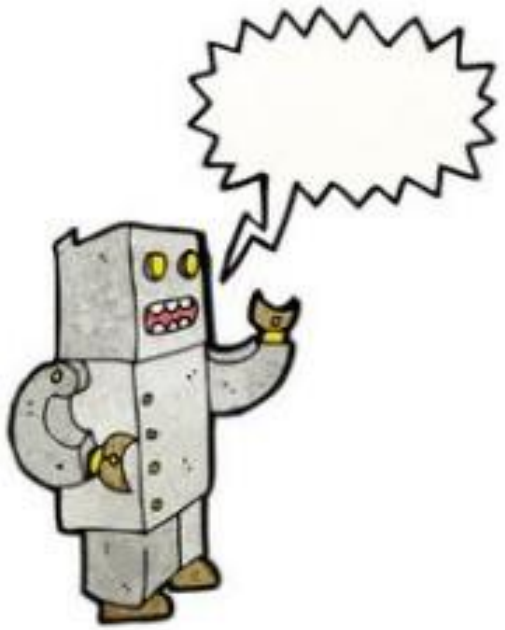
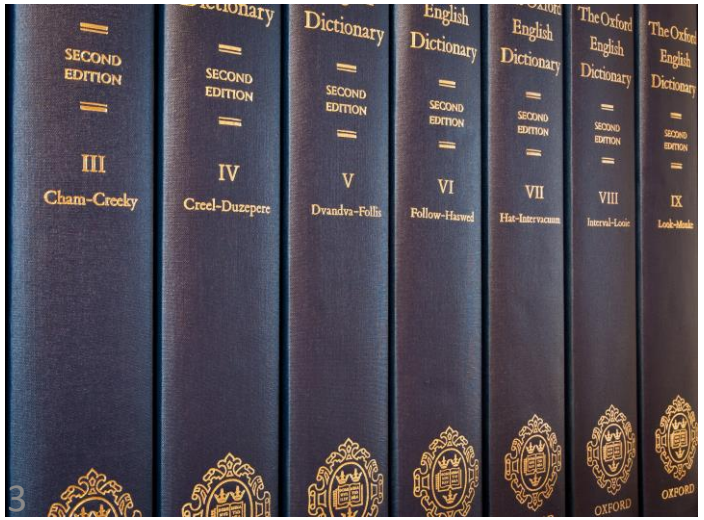
Natural Language Processing

- Complications of Real Natural Language
- Language models
- The bag-of-words model
- Naïve Bayes Classifier
- N-gram word models
- Part-of-speech (POS) tagging
- Natural Language Tasks



Complications of Real Natural Language

- The grammar of real English is endlessly complex
  - Quantification (“*Every agent feels a breeze.*”)
  - Pragmatics (context! “*You have a green light*”)
  - Long-distance dependencies
    - “*Who did the agent tell you to give the gold to?*”
  - Time and tense (“*Ali loves Bo*” vs. “*Ali loved Bo*”)
  - Ambiguity (“*Squad helps dog bite victim*”)
  - Lexical ambiguity (more than one meaning)



Processing Real Natural Language

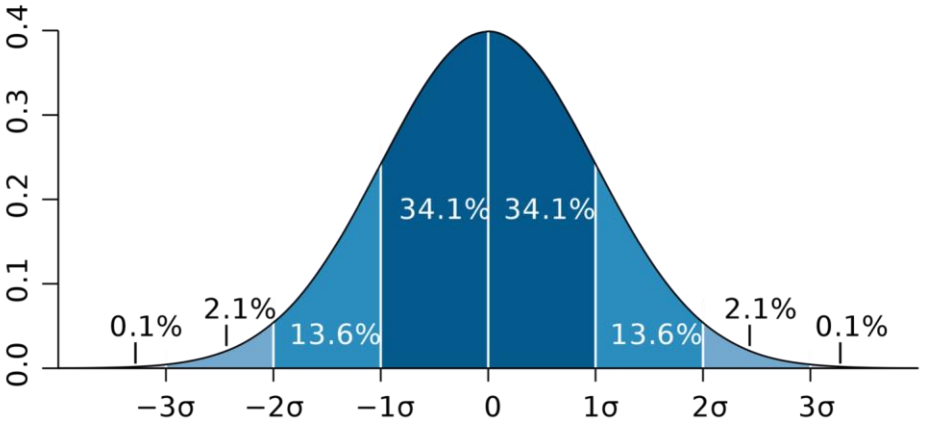
- Disambiguation
  - A process of recovering the most probable intended meaning of an utterance.
- The world model
  - The likelihood that a proposition occurs in the world
- The mental model
  - The likelihood that the speaker forms the intention of communicating a certain fact to the hearer.

Complications of Real Natural Language

- **The language model**
  - The **likelihood** that a certain string of words will be chosen, given that the speaker has the intention of communicating a certain fact.
- The acoustic model
  - for spoken communication, the likelihood that a particular sequence of sounds will be generated, given that the speaker has chosen a given string of words.

Language models

- A language model
  - A probability distribution describing the likelihood of any string.
- What is the likelihood?
  - “Do I dare disturb the universe?” vs. “Universe dare the I disturb do?”
- Language models are at the heart of a broad range of natural language tasks.



Language models

- With a language model, we can
  - predict what words are likely to come next in a text, and thereby suggest completions for an email or text message.
  - compute which alterations to a text would make it more probable, and thereby suggest spelling or grammar corrections.
  - compute the most probable translation of a sentence.
  - compute the most likely answer to a question with some example question/answer pairs as training data.

The bag-of-words language model

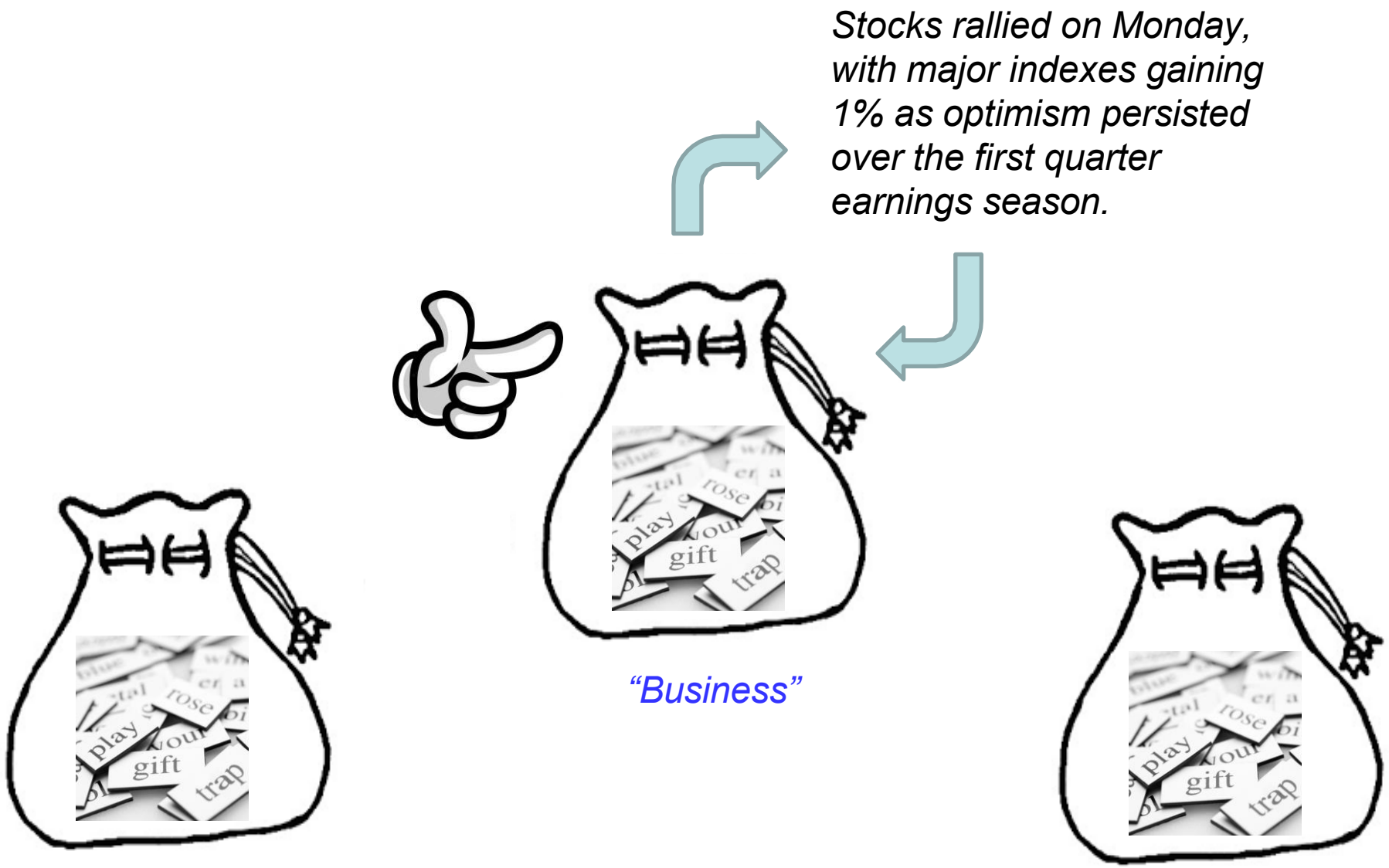
- How to classify a sentence into "*business*" or "*weather*" class?
  - "*Stocks rallied on Monday, with major indexes gaining 1% as optimism persisted over the first quarter earnings season.*"
  - "*Heavy rain continued to pound much of the east coast on Monday, with flood warnings issued in New York City and other locations.*"
- How can we **generate** a sentence like this?
  - Imagine that for each category (*business*, *weather*, etc.) we have a bag full of words
  - Imagine each word written on a slip of paper inside the bag
  - The more common the word, the more slips it is duplicated on

The bag-of-words language model

The bag-of-words language model

- To generate text
  - First select one of the bags and discard the others.
  - Reach into that bag and pull out a word at random; this will be the first word of the sentence.
  - Then put the word back and draw a second word.
  - Repeat until an end-of-sentence indicator (e.g., a period) is drawn.
- This leads to a **generative** model that describes a process for generating a sentence

The bag-of-words language model



Naïve Bayes Classifier

- The likelihood of a sentence is therefore
  - P(*"Stocks rallied … earnings season."* | *Business)* * *P(Business)*

    = P(*"Stocks"* | *Business)* * P(*"rallied"* | *Business)* *…*…*…

    * P(*"earnings"* | *Business)* * P(*"season."* | *Business)* * *P(Business)*
- Formally, this can be written as
  - $P(x_1, x_2, \ldots, x_n | y) * P(y)$

    $= P(x_1 | y) * P(x_2 | y) * \ldots * \ldots * \ldots * P(x_{n-1} | y) * P(x_n | y) * P(y)$
- Class-conditional independence

Naïve Bayes Classifier

- Bayes' formula

P(A,B) = P(A|B)P(B) = P(B|A)P(A)

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

A: "*Business" OR y*
B: "*Stocks rallied … earnings season." OR {*x_1, x_2, …, x_n*}*

P(A|B) = P(Business | "Stocks rallied … earnings season.")

OR

$= P(y \mid x_1, x_2, \ldots, x_n)$

Naïve Bayes Classifier

- Naïve Bayes Classifier for *k* classes

$$\boxed{P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}}$$

$$\hat{y} = \underset{k \in \{1,\ldots,K\}}{\operatorname{argmax}} \; p(C_k) \prod_{i=1}^{n} \underline{p(x_i \mid C_k)}$$

Estimated from training data

Naïve Bayes Classifier

- Training data: a corpus of text (labelled)
  - A corpus typically consists of at least a million words of text, and at least tens of thousands of distinct vocabulary words.
  - From a corpus we can estimate the prior probability of each category, $P(C_k)$, by counting how common each category is.
  - We can also use counts to estimate the conditional probability of each word given the category, $P(x_i|C_k)$
- Feature selection
  - We could drop words that are very rare, as well as words that are common to all classes.

N-gram word models

- The bag-of-words model has limitations
  - it falsely assumes that each word is **independent** of the others
  - "*first **quarter** earnings report*" vs "*fourth **quarter** touchdown passes*"
- We need a new model, where each word is dependent on previous words.
  - Considering dependence on ***all previous words*** in a sentence?
  - We can compromise to consider only the ***dependence between n adjacent words***.
  - This is known as an **n-gram model.**

N-gram word models

Joint probability of n words, $w_1, w_2, \ldots, w_n$

$$P(w_{1:n}) = P(w_1)P(w_2|w_1)P(w_3|w_{1:2})\ldots P(w_n|w_{1:n-1})$$
$$= \prod_{k=1}^{n} P(w_k|w_{1:k-1})$$

Now we only consider the dependence of a word on its previous N words

$$P(w_n|w_{1:n-1}) \approx P(w_n|w_{n-N+1:n-1})$$

Now the joint probability of n words reduces to a N-gram model

$$P(w_{1:n}) \approx \prod_{k=1}^{n} P(w_k|w_{k-1})$$

bi-gram model (N=2)

N-gram word models

- Unigram (i.e., bag-of-words), bigram, trigram, and 4-gram models are built to randomly sample word sequences:

n=1	*“logical are as are confusion a may right tries agent goal the was”*
n=2	*“systems are very similar computational approach would be represented”*
n=3	*“planning and scheduling are integrated the success of naive Bayes model is”*
n=4	*“taking advantage of the structure of Bayesian networks and developed various languages for writing "templates” with logical variables, from which large networks could be constructed automatically for each problem instance”*

Natural Language Processing

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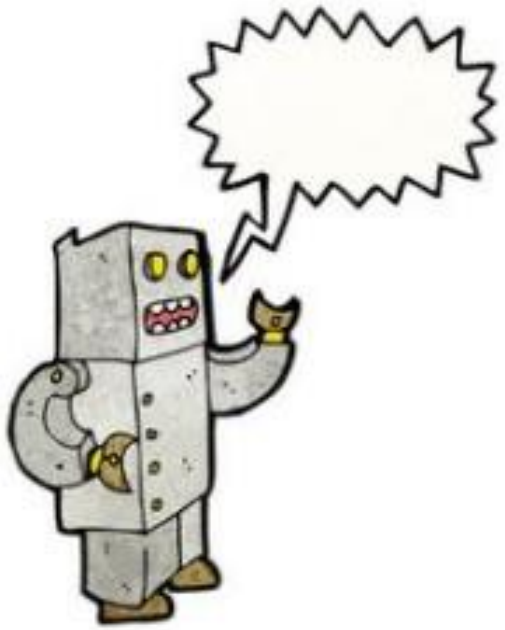
- Natural Language Tasks



Part-of-speech (POS) tagging

- Part of speech (POS)
  - A category of words that have similar grammatical properties
  - *noun, verb, adjective, adverb, pronoun, preposition, conjunction, interjec tion, numeral, article,* or *determiner*. (45 categories)
- POS tagging
  - A process to categorise words in a text with respect to part of speech, based on the definition of the word and its context.

heat	*water*	*in*	*a*	*large*	*vessel*
(verb)	(noun)	(prep)	(det.)	(Adj.)	(noun)

Part-of-speech (POS) tagging

- Why need POS tagging?
  - allow language models to capture generalizations such as “adjectives generally come before nouns in English.”
  - Named Entity Recognition
  - Information extraction patterns
  - Text-to-speech synthesis (the pronunciation of “record”)
  - …

TITLE:	Personalities - Senator Gareth Evans & Mr Carl-August Fleischauer, Under Secretary General, Legal Affairs, UN, 1991
BARCODE:	R11876218_0001
YEAR:	1991
IMAGE NUMBER:	A6135:K18/11/91/12

Part-of-speech (POS) tagging

- How do we do POS tagging?
  - We can solve it as a **classification** problem!
  - For POS tagging we would build a multi-class classifier
  - One for each part of speech (i.e., a category/class)
  - Ask this model how probable it is that a word is a member of that category, given the feature values for that word in its particular context.
  - (Let's look at the next slide first – can you recall it?)
  - Then, what are the feature values *f(x)*?

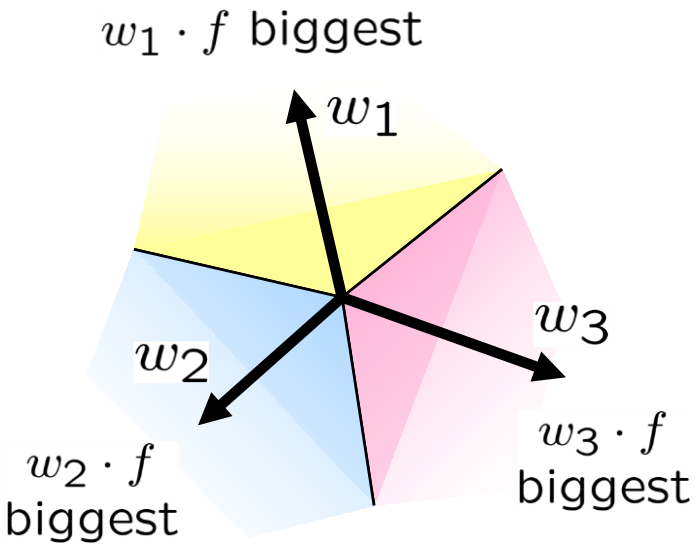
Multiclass Logistic Regression

Multi-class linear classification

A weight vector for each class: w_y

Score (activation) of a class y: $w_y \cdot f(x)$

Prediction w/highest score wins: $y = \arg\max_y \; w_y \cdot f(x)$



How to make the scores into probabilities?

$$\underbrace{z_1, z_2, z_3}_{\text{original activations}} \rightarrow \underbrace{\frac{e^{z_1}}{e^{z_1}+e^{z_2}+e^{z_3}}, \frac{e^{z_2}}{e^{z_1}+e^{z_2}+e^{z_3}}, \frac{e^{z_3}}{e^{z_1}+e^{z_2}+e^{z_3}}}_{\text{softmax activations}}$$

Part-of-speech (POS) tagging

- Then, what are the feature values *f(x)*?
  - Typically use binary-valued features that encode information about the word being tagged and perhaps other nearby words,
  - as well as the category that was assigned to the previous word and perhaps the category of earlier words.
  - Features can depend on the exact identity of a word, some aspects of the way it is spelled, or some attribute from a dictionary entry.
- Logistic regression itself does not have the notion of a sequence of inputs (although there are ways to handle this issue)

Generative and Discriminative Models

- Logistic regression is a **discriminative** model.
  - It learns a conditional probability distribution P(C | W), meaning that it can assign categories given a sequence of words, but it can't generate random sentences.
- Naive Bayes models are **generative** models
  - They learn a joint probability distribution, P(W, C) (i.e., P(W|C)P(C))
  - We can generate a random sentence by sampling from that probability distribution to get a first word (with category) of the sentence, and then adding words one at a time.

Generative and Discriminative Models

- Discriminative models usually have a lower error rate
  - They model the intended output directly
  - They make it easier for an analyst to create additional features.

- Generative models tend to converge more quickly
  - May be preferred when the available training time is short, or when there is limited training data.

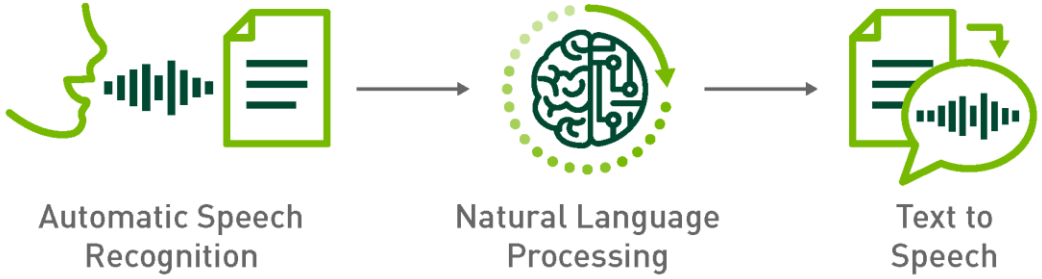
$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Natural Language Tasks

- Speech recognition
  - The task of transforming spoken sound into text.
  - Upon it, we can then perform further tasks (such as question answering) on the resulting text.
  - Current systems have a word error rate of about 3% to 5% (depending on details of the test set), similar to human transcribers.

Natural Language Tasks

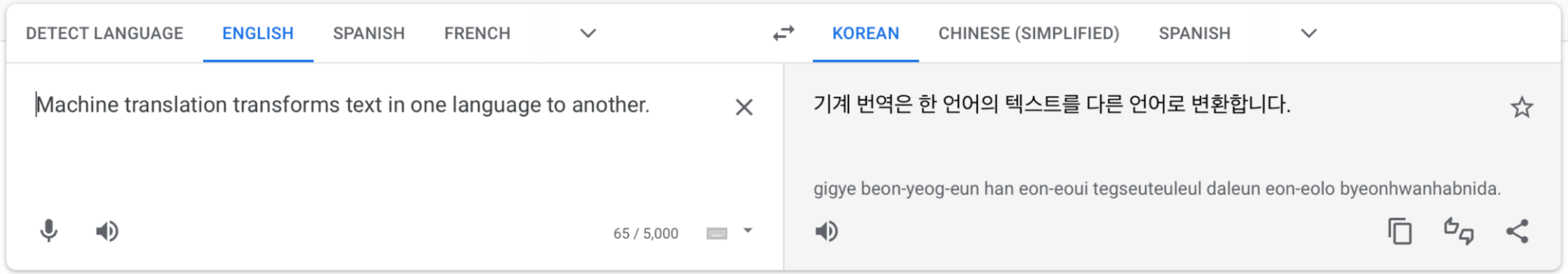
- Text-to-speech synthesis
  - The inverse process—going from text to sound.
  - The challenge is to pronounce each word correctly, and to make the flow of each sentence seem natural, with the right pauses and emphasis.
- Synthesizing different voices
  - A generic male or female voice
  - Allowing for regional dialects
  - Even imitating celebrity voices.

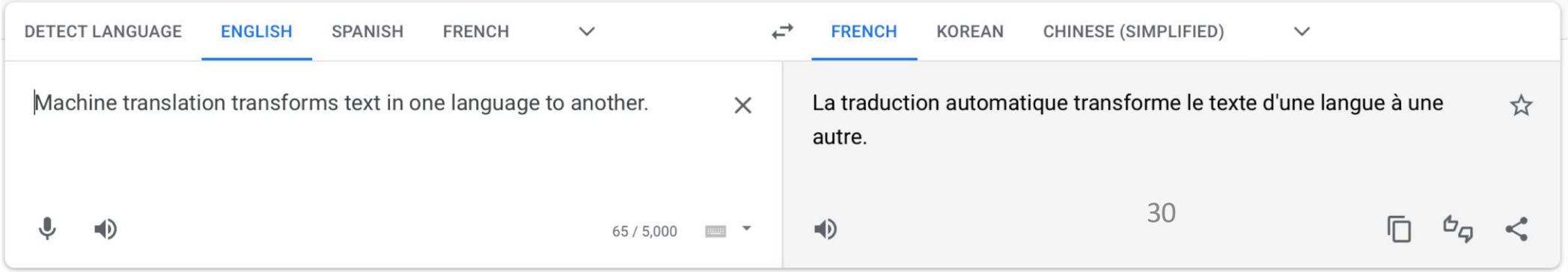


Natural Language Tasks

- Machine translation
  - Transforms text in one language to another.
  - Systems are usually trained using a bilingual corpus: a set of paired documents, where one member of the pair is in, say, English, and the other is in, say, French.
  - Early days: n-gram models
  - Nowadays: Recurrent neural sequence-to-sequence models, and transformer models

Natural Language Tasks



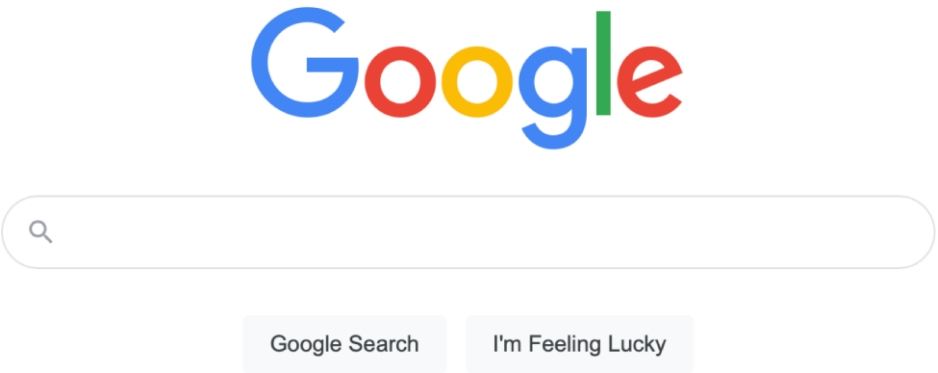


Natural Language Tasks

- Information extraction
  - The process of acquiring knowledge by skimming a text and looking for occurrences of particular classes of objects and for relationships among them.
  - A typical task is to extract instances of addresses from Web pages, with database fields for street, city, state, and zip code;
  - or instances of storms from weather reports, with fields for temperature, wind speed, and precipitation.
  - It is hard if the text is not structured and we are trying to extract all facts, rather than a specific type.

Natural Language Tasks

- Information retrieval
  - The task of finding documents that are relevant and important for a given query.
  - Internet search engines such as Google and Baidu perform this task billions of times a day.



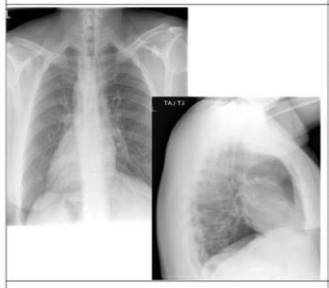
Natural Language Tasks

- Question Answering
  - The query is a question and the response is an actual answer.
  - Used Web information retrieval to radically increase their breadth of coverage
- An example
  - Query: "*Who is the first prime minister of Australia?*"
  - Answer: "*Edmund Barton was the first Prime Minister of Australia. He held office from 1901 to 1903.*"

Natural Language Tasks

■ Visual Question Answering (VQA): Image + Text

Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering (CVPR 2017)

Input Images	Ground Truth	Ours
	**Impression:** No acute cardiopulmonary finding. **Findings:** **The heart size and cardio mediastinal silhouette are normal.** There is no focal air space opacity, pleural effusion, or pneumothorax. **The osseous structures are intact with mild degenerative changes in thoracic spine**	**Impression:** no acute cardiopulmonary finding . **Findings:** **the cardio mediastinal silhouette is normal in size and contour** . the lungs are clear . there is no pleural effusion or pneumothorax . **the osseous structures are intact with mild degenerative changes in thoracic spine**
	The thorax is symmetrical, mediastinal heart shadow is in the middle, no enlarged lymph nodes are found in mediastinum, **bilateral lung markings are enhanced, patchy ground glass shadows are seen in both lungs, the boundary is fuzzy, some density is uneven**, lobar bronchus is unobstructed, and no abnormal density shadow is found in bilateral thoracic cavity	The chest is symmetrical, mediastinal heart shadow is in the middle, no enlarged lymph node shadow is found in mediastinum, **bilateral lung markings are enhanced, multiple flake ground glass shadows are seen in both lungs, the boundary is fuzzy, the bilateral hilar is not big, some of the density is uneven**, the lobar bronchus are unobstructed, and no abnormal density shadow is found in bilateral chest.

A Self-Boosting Framework for Automated Radiographic Report Generation (CVPR 2021)

Natural Language Tasks

Text → label (classification)

- Sentiment / emotion
- Topic / intent
- Toxicity / safety / spam
- Stance, sarcasm
- Language ID, genre/style

Text → spans/tags (sequence labeling)

- Part-of-speech tagging
- Named entity recognition (NER)
- Chunking (noun/verb phrases)
- Slot filling
- Keyphrase extraction

Text → structure (linguistic analysis)

- Dependency / constituency parsing
- Coreference resolution
- Semantic role labeling
- Word sense disambiguation
- Entity linking (mention → KB entity)

Text → facts/tuples (information extraction)

- Relation extraction (e.g., *works_for(person, org)*)
- Event extraction (event type + arguments)
- OpenIE (subject–predicate–object triples)

Text ↔ text (semantic matching / inference)

- Semantic textual similarity (STS)
- Paraphrase / duplicate detection
- Natural language inference (entail/contradict/neutral)
- Retrieval and reranking (query–document matching)

Text → text (generation / transformation)

- Machine translation
- Summarization
- Question answering (extractive or generative)
- Dialogue / chat
- Paraphrasing / rewriting
- Simplification
- Style transfer
- Data-to-text (tables/records → narrative)

Speech ↔ text (often included under "NL" tasks)

- Speech recognition (ASR): speech → text
- Text-to-speech (TTS): text → speech
- Spoken language understanding: speech/text → intent + slots
- Speech translation

Summary

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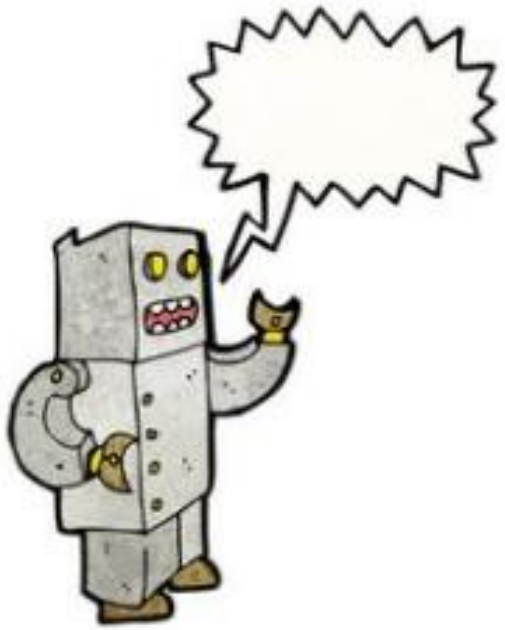

Acknowledgement

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Questions?

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